

Linear Regression from Scratch

Predicting CO₂ Emissions with Pure Mathematics

Project 23 — Linear Algebra for Computer Science

German International University

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Year: 2025/2026

Why CO₂ Emissions Matter

Transport accounts for a major share of global emissions; predicting vehicle CO₂ supports sustainability decisions.



Emissions depend on engine characteristics



Prediction enables policy/standards, consumer awareness, and fleet planning

Vehicle specs

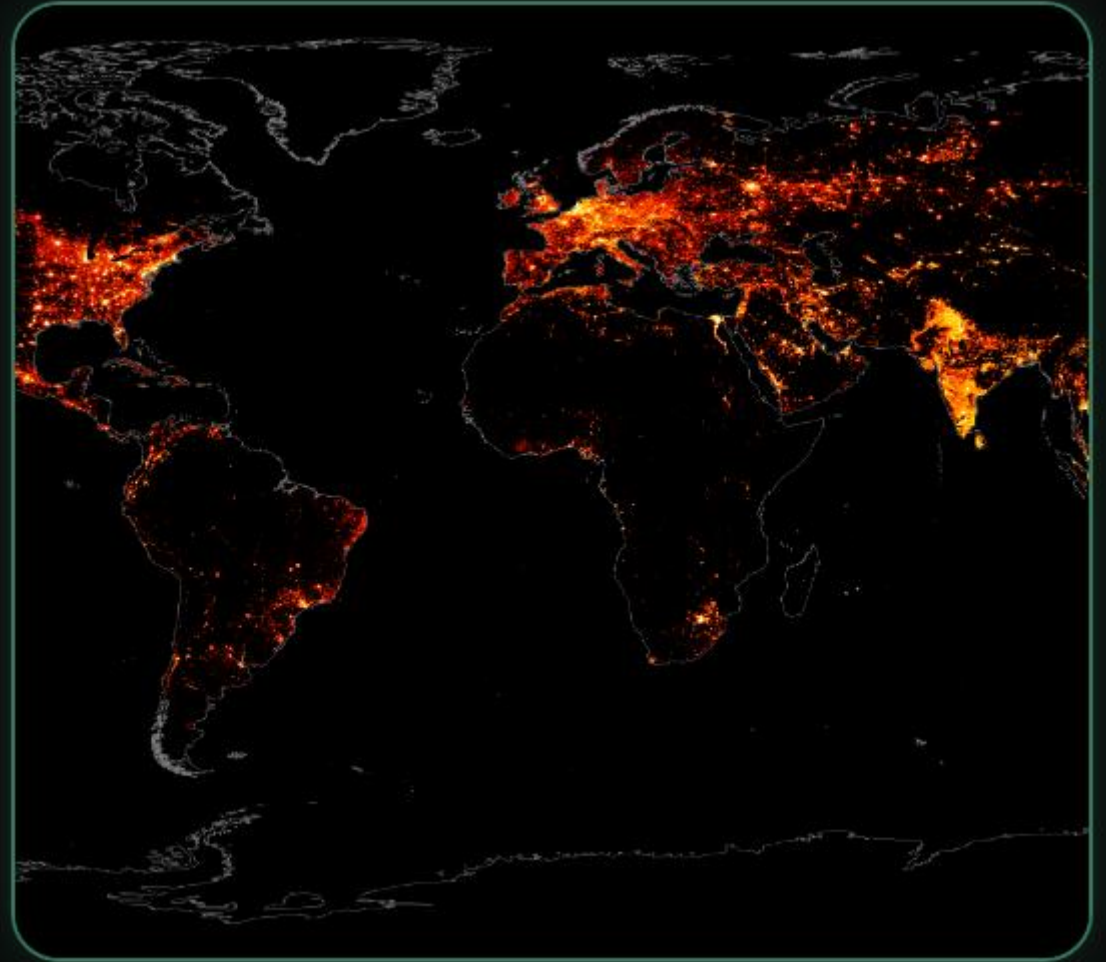


CO₂ prediction model



Decision impact

(policy / product /
consumer)



Real-World Data: Canadian Vehicles



Dataset

~7,385 vehicle records

X

Features

Engine Size (L), Cylinders

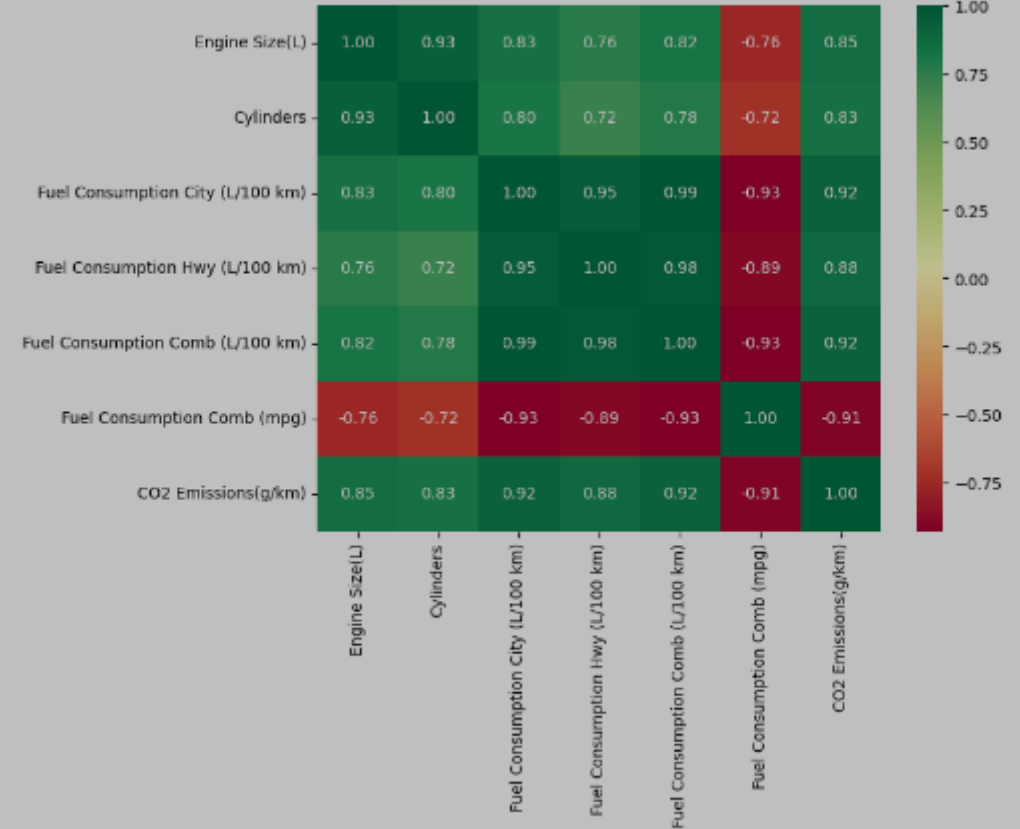
y

Target

CO₂ emissions (g/km)

Two intuitive features can explain a meaningful portion of emission variance.

Correlation Matrix (CO₂ Emissions Dataset)



Linear Algebra Foundations



Vectors & Matrices

Representing data as X (features), y (targets), and w (weights).



Minimizing Error

Finding the optimal weights to reduce the difference between predictions and actual values.



Least Squares Concept

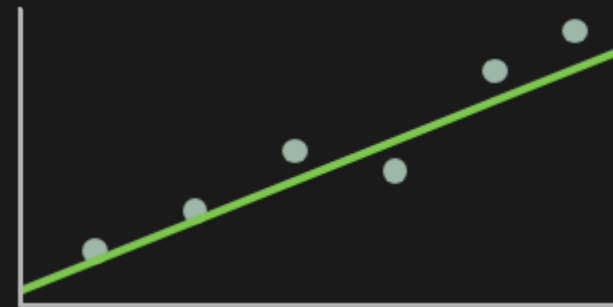
Minimizing the sum of squared residuals to find the best-fitting line.

Matrix Representation

$$y = X w + b$$

Target = Features $\times$ Weights + Bias

Data $\rightarrow$ Line Fit



Normal Equation



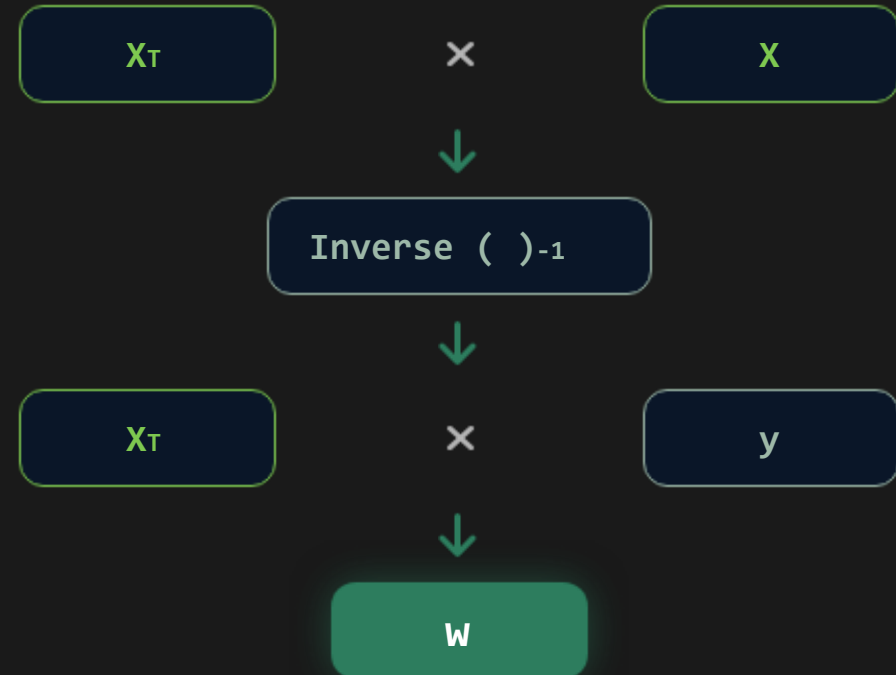
Direct closed-form solution

Compute once, get optimal weights.

Formula

$$W = (X^T X)^{-1} X^T y$$

Matrix Multiplication Flow



Gradient Descent

Formula

$$W = W - \alpha \cdot X^T(XW - y)$$



Iterative Optimization

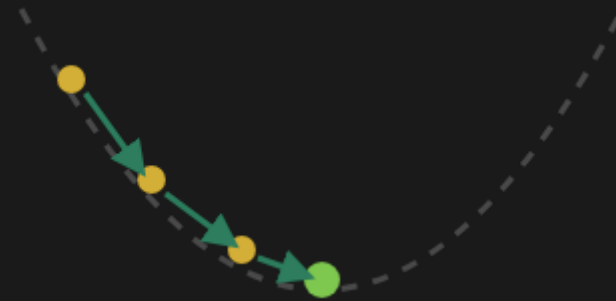
Gradually updates weights step-by-step to find the minimum error.



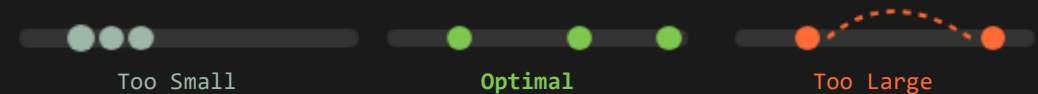
Key Parameters

Requires tuning the **learning rate (α)** and setting the number of iterations for convergence.

Step-by-Step Descent



Learning Rate (α) Impact



Ridge Regression



L2-Regularized Solution

Adds a penalty term to the loss function to constrain weight magnitudes and improve model stability.

Formula

$$W = (X^T X + \lambda I)^{-1} X^T y$$



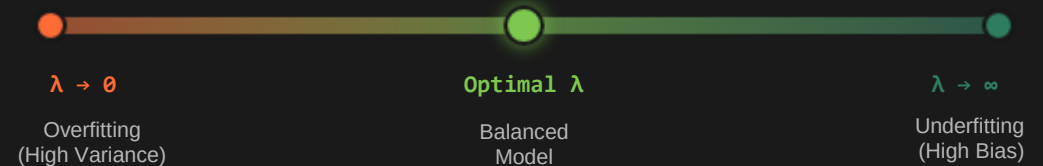
Prevents Overfitting

The regularization parameter λ controls the trade-off between fitting training data and keeping weights small.

Regularization Effect



λ Tuning Impact



Engineering the Solution

Tools



NumPy

(core math)



pandas

(data handling)



matplotlib

(plots)

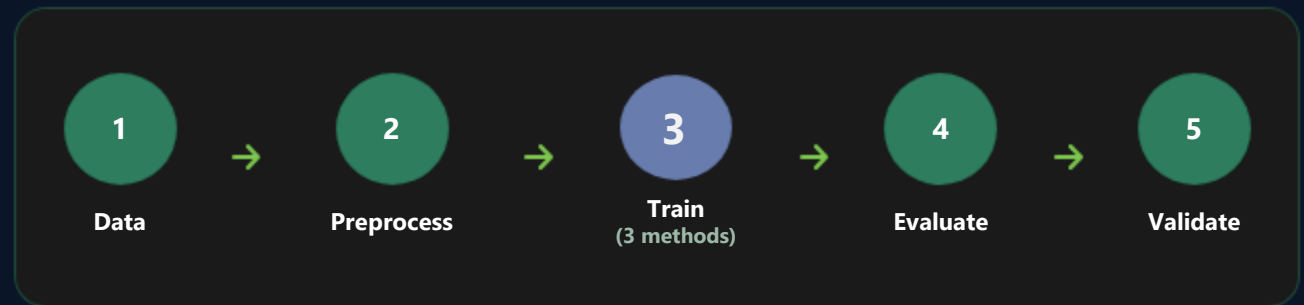


sklearn

(validation only)

Key decisions

- Prefer `np.linalg.solve()` over explicit inverse
- Normalize features for gradient descent
- Exclude bias term from Ridge regularization



All Methods Agree

R² Score

≈ 0.72–0.73

MSE

≈ 900

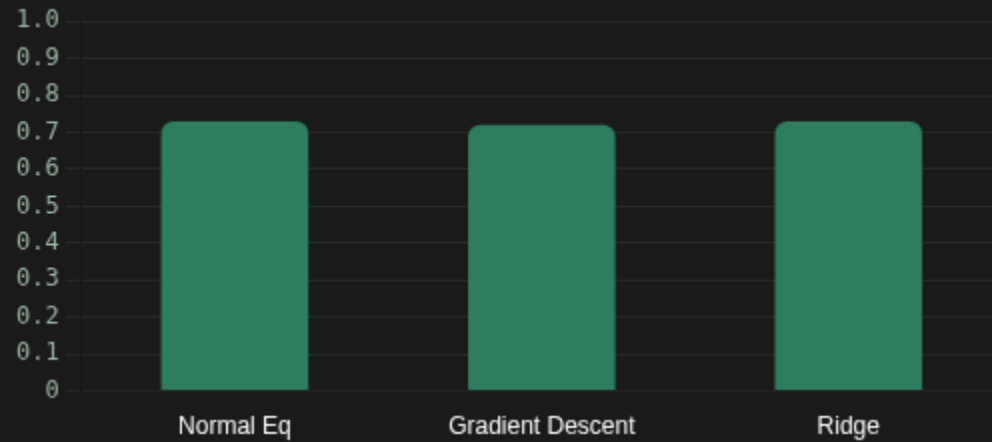
Predictions within

~30 g/km

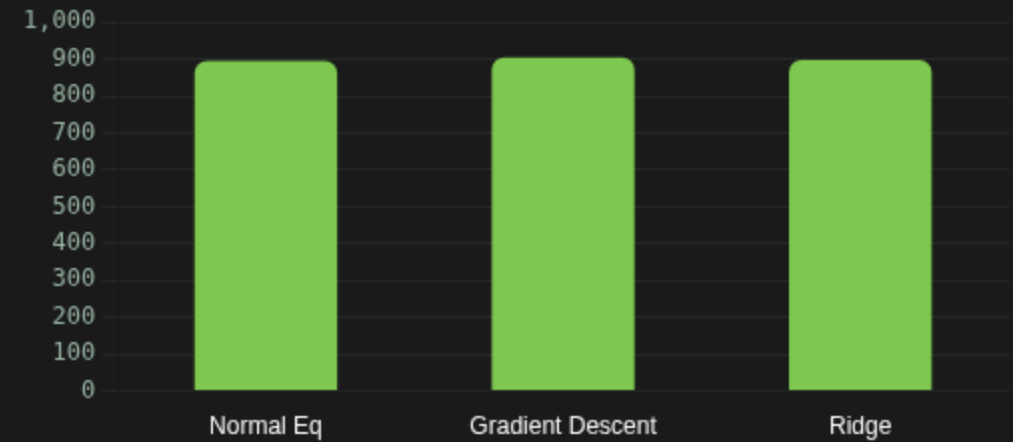
Weight diff vs sklearn

≈ 10⁻¹²

R² by Method



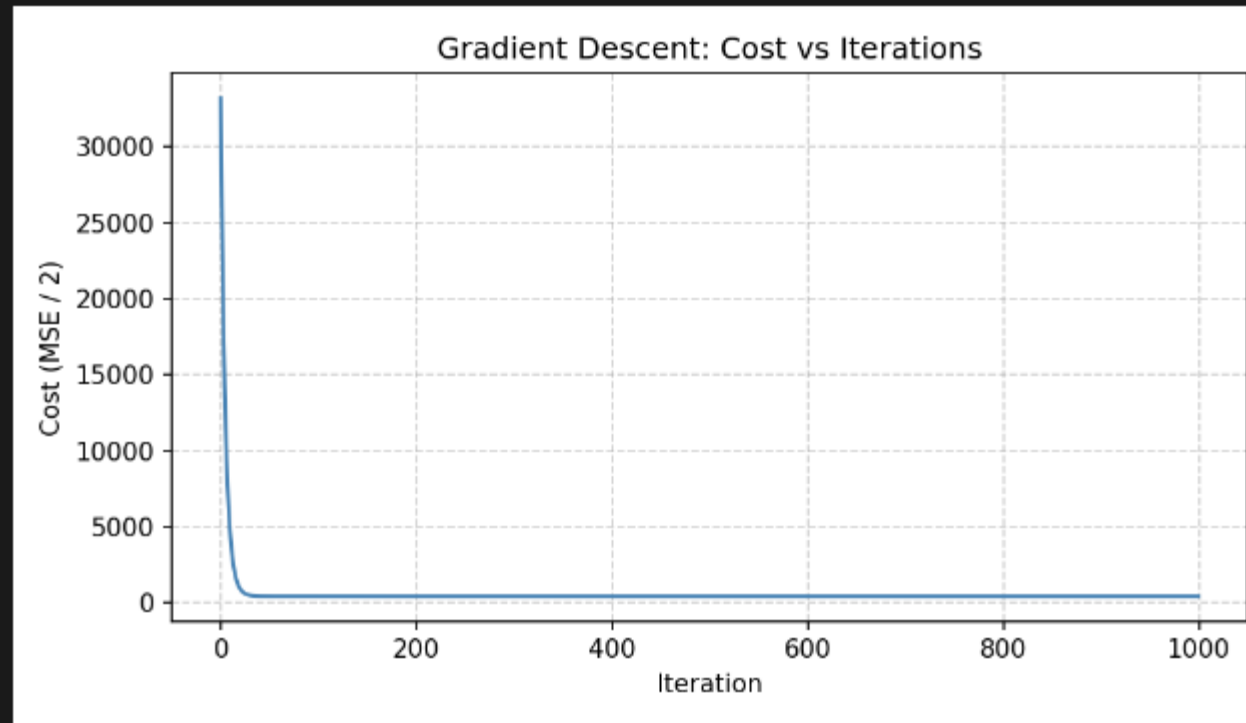
MSE by Method



Watching the Model Learn

Rapid error drop, then stable convergence.

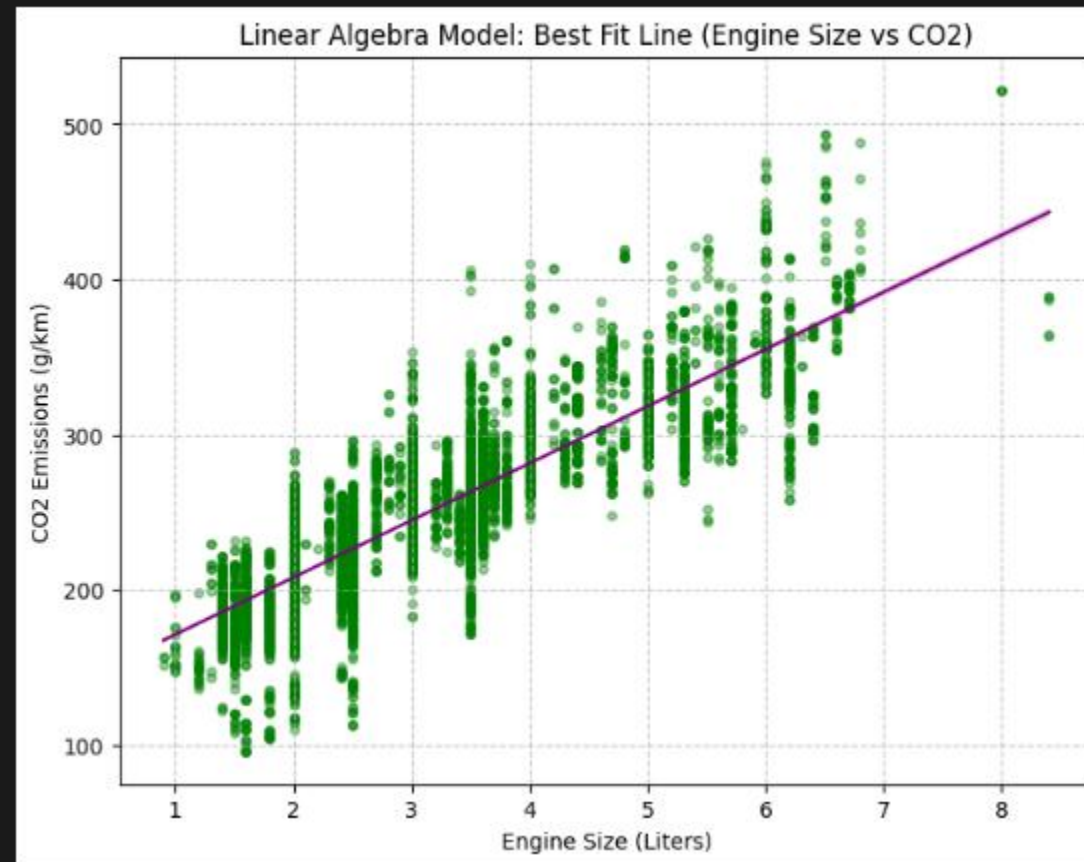
● Rapid learning phase



● Convergence plateau

From Math to Reality

- Insight: A simple linear model produces practical accuracy ($\approx \pm 30$ g/km).



What We Learned



Mathematical Truth

Linear algebra fundamentals are sufficient to build regression end-to-end



Practical Insight

From-scratch weights match sklearn ($\approx 10^{-12}$ difference)



Real-World Impact

Emission predictions support sustainability decisions



Green AI impact

Thank You

Questions?

 Ahmed Roshdy

 Omar Shaker

 Mark Tamer

 Carlos Emad

 Khalid Osam

 Yousef Hany

 Hassan Eslam

 Mohamed Elsayed

 Ahmed Fouad

